

Nanoscopic cross-grain cation homogenization in perovskite solar cells

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Multiscale cation inhomogeneity has been a major hurdle in state-of-the-art formamidinium–caesium (FA–Cs) mixed-cation perovskites for achieving perovskite solar cells with optimal power conversion efficiencies and durability. Although the field has attempted to homogenize the overall distributions of FA–Cs in perovskite films from both plan and cross-sectional views, our understanding of grain-to-grain cation inhomogeneity and ability to tailor it—that is, spatially resolving the FA–Cs compositional difference between individual grains down to the nanoscale—are lacking. Here we reveal that as fundamental building blocks of a perovskite film, individual grains exhibit cationic compositions deviating from the prescribed ideal composition, severely limiting the interfacial optoelectronic properties and perovskite layer durability. This performance-limiting nanoscopic factor is linked to thermodynamic-driven morphological grooving, leading to a segmented surface landscape. At the grain triple junctions, grooves form nanoscale groove traps that hinder the mixing of solid-state cations across grains and thus retard inter-grain FA–Cs mixing. By rationally modulating the heterointerfacial energies, we reduced the depth of these nanoscale groove traps by a factor of three, significantly improving cation homogeneity. Perovskite solar cells with shallower nanoscale groove traps demonstrate enhanced power conversion efficiencies (25.62%) and improved stability under various standardized international protocols. Our work highlights the significance of resolving surface nano-morphologies for homogeneous properties of perovskites.

The recent efforts to engineer the compositions, microstructures and phases in perovskite thin films have led to rapidly climbing certified power conversion efficiencies (PCEs) and promising stability parameters in perovskite solar cells (PSCs)^{1–4}. State-of-the-art PSCs invariably adopt formamidinium lead triiodide (FAPbI₃) based on perovskite absorbers owing to their relatively ideal bandgaps^{5–7}. However, pure-composition FAPbI₃ perovskite undergoes a facile, detrimental phase transition to nonperovskite^{8,9}. Therefore, a certain amount of caesium (Cs) cations are generally incorporated into FAPbI₃ to attain more stable crystal structures. The resultant

formamidinium–caesium (FA–Cs) mixed-cation perovskites have become the key contributors to best-performing PSCs^{10–13}. Nevertheless, achieving a completely homogeneous FA–Cs mixture in perovskite thin films is challenging^{14–16}, because the film fabrication is generally performed in rapid, non-equilibrium solution conditions. Although, in principle, the FA–Cs cationic mixing in crystalline perovskite lattices is entropy-driven and thermodynamically favoured^{10,11}, an ideal FA–Cs cation homogeneity at the nano-, micro- and macro-scales can be retarded by various kinds of microstructural disorders^{15,17}. The cation inhomogeneity inevitably creates a range of FA–Cs perovskite phases

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contained in films while deviating from the prescribed ideal composition, leading to unfavoured optoelectronic properties and reduced multi-factor stabilities^{10,11}. The reported studies to address this issue mainly focused on tackling the FA–Cs inhomogeneity without spatially resolving the differences between individual grains while demonstrating the advantages of microscopically and macroscopically attaining a higher degree of FA–Cs homogeneity^{14,15,17,18}. Among these, Liang et al.¹⁴ and Yang et al.¹⁵ have revealed and mitigated out-of-plane and in-plane FA–Cs cation inhomogeneities in the context of the entire perovskite thin films, respectively, leading to impressive performance enhancements. Nevertheless, the understanding and tailoring of grain-to-grain cation inhomogeneity down to the nanoscale remains elusive. Because individual grains are fundamental building blocks for perovskite thin films, eliminating grain-to-grain cationic variations can be an effective path towards a homogeneous, multiscale FA–Cs composition landscape for the entire PSCs.

In the crystallization of a typical FA–Cs perovskite thin film, local heterogeneous solution environments inevitably result in variations of initial perovskite compositions between as-formed grains. Such grain-to-grain variation in FA–Cs ratios is expected to be partly alleviated once all grains coalesce with grain boundaries (GBs) formed in the later stage of film growth. This work reveals that the degree of final grain-to-grain FA–Cs homogeneity is strongly linked to the thermodynamics-driven morphologic grooving of as-formed GBs^{19–21}. The resultant grooves at the grain triple junctions, where three profiles of GBs merge, geometrically create nanoscale trap-like morphologies on the perovskite film surface, detrimentally influencing the solid-state cation exchange between adjacent grains. To address this issue, we leverage the effect of the perovskite heterointerfacial chemistry on the geometric characteristics of such nanoscale groove traps (nano-GTs). By pre-incorporating a rationally selected butylammonium acetate (BAAC) additive at the film bottom, we reduced the nano-GT depths by a factor of three, unlocking an unprecedented cross-grain cation homogenization. Minimum grain-to-grain FA–Cs heterogeneity is observed for resultant perovskite films, leading to beneficial surface electronic energetics and, thus, improved heterointerfacial carrier dynamics. It also contributes to high film stability and interfacial mechanical reliability. On the basis of the grain-resolved fundamental understandings of FA–Cs heterogeneity, n–i–p structured PSCs with shallowed nano-GTs deliver PCEs up to 25.62%, as well as high multi-factor stabilities under various standardized international protocols (ISOS-L-1I maximum-power-point tracking test, IEC 61215:2021 MQT 11 thermal cycling test and IEC 61215:2021 MQT 13 damp heat test), attesting to the mechanistic effects of nano-GT geometry-dependent cross-grain cation homogenization. Different from reported studies on buried interfaces^{22,23}, this work establishes a direct link between surface nano-morphologies and cation distribution in perovskites. It also clarifies the elusive role of grain-to-grain FA–Cs homogeneity in film properties and device performance, guiding processing innovation for more efficient and stable PSCs.

Cross-grain microstructure featuring nano-groove traps

The formation of perovskite films occurs through rapid nucleation and crystallization from solution, which inherently results in a polycrystalline microstructure featured by numerous GBs²⁴. Since in PSCs, the surface will dominate owing to the nanoscale thickness of the perovskite films, most GBs are expected to emerge onto the surface, which causes morphological grooving at the intersection of GBs and the surface^{21,25}. Owing to the intrinsic microstructure of the GB network, these grooves tend to merge at grain triple junctions, as schematically illustrated in Fig. 1a, creating trap-like nanostructures on the film surface^{21,25}. The geometry of nano-GTs is determined by line tension equilibrium at the interaction point of three intersecting groove roots^{20,26,27}:

$$\frac{\gamma_{\text{TL}}}{\gamma_{\text{IS}}} = 3 \sin \zeta \quad (1)$$

where γ_{IS} and γ_{TL} are groove root line tension energy and triple line tension energy of nano-GTs, respectively. ζ is the angle between the groove root and the horizontal film plane. Here, ζ for the three groove roots are assumed to be identical. The energy profiles and their relations for nano-GTs are schematically illustrated at the bottom of Fig. 1a. For the given perovskite precursor composition and deposition used in the fabrication, γ_{TL} can be considered constant. Therefore, increasing γ_{IS} will reduce groove root angle (ζ), resulting in a shallowing effect. The variation of γ_{IS} will follow the same trend as that of perovskite surface energy, according to equation (2) in Methods. We then derived the expression for the depth of nano-GT (d_{GT}) that elaborates its quantitative relation with ζ (equations (3)–(5) in Methods). In brief, we found that a lower groove root angle (ζ) leads to a smaller nano-GT depth (d_{GT}), that is, shallower nano-GTs.

The above theory guided us to experimentally use a volatile chemical additive, BAAC, to induce a geometric change of nano-GTs. We prepared two FA–Cs perovskite film samples, one with pre-incorporating BAAC into the SnO_2 electron transport layer (ETL) and the other without BAAC, denoted as the target and pristine films, respectively. Note that for the materials experiments, the stable $\text{FA}_{0.9}\text{Cs}_{0.1}\text{PbI}_3$ composition is used. BAAC is expected to be released from the ETL upon the sequential annealing to influence the bottom surface of the perovskite, and it causes morphological effects when the intimate contact between the BAAC pre-incorporating ETL and perovskite bottom starts to form (see Supplementary Fig. 1 for more discussion). We then mechanically peeled off both perovskite films from the ETL and performed high-resolution atomic force microscopy (AFM) scans on the flipped perovskite film bottoms. We confirmed that the delamination causes no notable changes in grains and GBs (Supplementary Fig. 2). As seen in Fig. 1b,c and Supplementary Fig. 3, the AFM topography of the pristine film shows a significant height contrast between grains and GBs, which is clearly reduced in the target case. Figure 1d,e shows the local AFM images focusing on single typical nano-GTs in both films, further attesting to the shallower nano-GTs after the BAAC addition.

To extract the quantitative information regarding the nano-GT shallowing effect, we acquired nano-GTs' AFM height profiles and compared the d_{GT} distributions for both films (Fig. 1f). In the target film, the mean d_{GT} decreases from 15.3 nm to 4.4 nm after BAAC-assisted nano-GTs tailoring. We further performed first-principles calculations based on the density functional theory (DFT) to gain an understanding of the impact of BAAC chemical modification on perovskite heterointerface energies. The calculation results (Fig. 1g) reveal that when the perovskite surface is modified with BAAC, the relative bottom surface energy (γ_{S}) is increased to 1.068 eV nm^{−2} compared with the pristine (set as 0 eV nm^{−2}). Meanwhile, the groove line tension energy (γ_{IS}) in each determined nano-GT is positively correlated with γ_{S} , as elaborated in Methods. In this context, the γ_{IS} , because of the BAAC modification, prompts a decrease in the groove root angle (ζ) and depth (d_{GT}), consistent with equations (1) and (5). Further insights into the chemical role and selection rationale of BAAC are provided in Supplementary Figs. 4–7 and Supplementary Note 1. It is worth noting that quadruple junctions are occasionally observed in the film, which behave similarly to the triple junctions (Supplementary Figs. 8–10 and Supplementary Note 2).

Nano-groove trap depth-dependent cation homogeneity

We then conducted cathodoluminescence (CL) microscopy to hyperspectrally image the delaminated perovskite bottom surfaces in both films. CL microscopy is a well-established method for quantitatively probing semiconductor materials at the nanoscale, allowing localized excitation to assess the impact of grain microstructure on excited-state

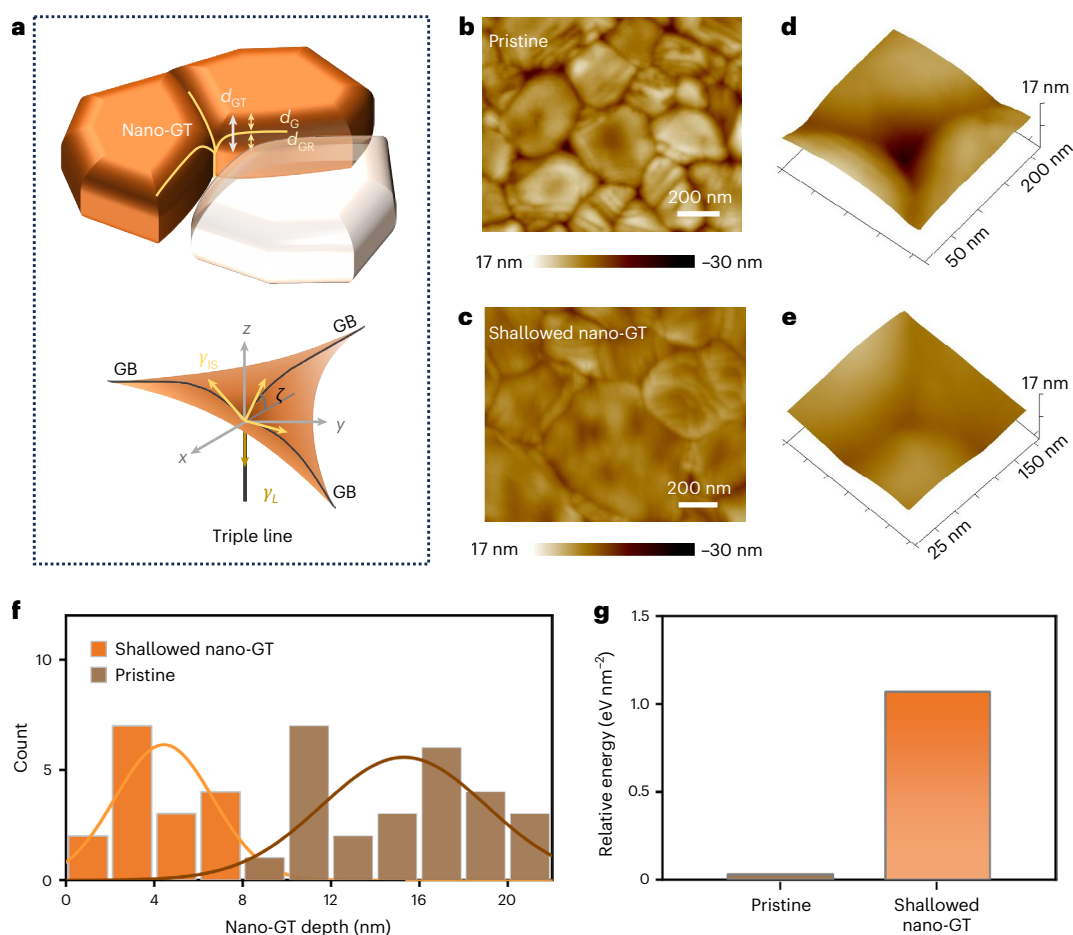


Fig. 1 | Cross-grain microstructures. **a**, Schematic illustration of a nano-GT in a typical polycrystalline perovskite film, the 3D view of the line tension equilibrium at the nano-GT, as well as the energy relationship at nano-GT. d_{GT} , d_G and d_{GR} represent the depth of nano-GT, grain-boundary groove and the groove root, respectively. γ_{IS} and γ_{TL} are the groove root line tension energy and the triple line tension energy of the nano-GTs, respectively. ζ is the angle between the groove root and the horizontal film plane. **b, c**, AFM images comparing the microstructure of a pristine perovskite film (**b**) with that of a shallowed nano-GT film (**c**). For both samples, the AFM-characterized film surface is the buried

surface in contact with the SnO₂-ETL-coated ITO substrate. **d, e**, High-resolution AFM images in 3D view of the nano-GTs in pristine (**d**) and shallowed nano-GT (**e**) perovskite films. **f**, The comparison of the statistical distributions of d_{GT} of the pristine sample and the shallowed nano-GT films. The values of d_{GT} are acquired from 17 measurements for the shallowed nano-GT film and 26 measurements for the pristine film, respectively. **g**, A comparison of the relative interfacial energy of the perovskite–SnO₂ interface of pristine SnO₂ and BAAC–SnO₂ via DFT calculations.

energetics and dynamics^{28–31}. Supplementary Fig. 11 schematically illustrates the methodology, showing the correlative acquisition of both the scanning electron microscope (SEM) microstructural information and the local CL spectral feature. Herein we observed a positive effect of nano-GT shallowing on the initial perovskite phase purity. In pristine films, there is a random distribution of rod-shaped grains in the polycrystalline perovskite matrix, identified as δ -CsPbI₃ from their CL emission at 443 nm (ref. 32). But δ -CsPbI₃ is absent in the target film, where the CL spectrum shows a symmetric peak at 810 nm, corresponding to the pure FA–Cs perovskite^{33–35} (Supplementary Figs. 12 and 13). The space-averaged CL spectra for both films further reveal notable shape asymmetry for the pristine case (Supplementary Fig. 14a), attributed to the FA–Cs cation inhomogeneity. Local CL emission fitting across the hyperspectral datasets was performed to visualize cation inhomogeneity at the grain-to-grain level^{32,36}. Figure 2a–f shows correlated SEM images together with the maps of CL emission peak wavelength for both perovskite films, which are spatially resolved by the automated emission fitting function. In the pristine film, the variation in the CL emission wavelengths is observed among different grains (Fig. 2d). We chose three representative locations from different grains and extracted their CL spectra. We observed noticeable peak shifts, with

2–3 nm differences in wavelength, across the three locations (Fig. 2g). This unravels the grain-to-grain FA–Cs cation inhomogeneity in the pristine film. By contrast, by doing the same analysis in the target film, we found no discernible change in CL emission peak wavelengths for all the examined grains (Fig. 2f,i), and the CL peak in the averaged spectrum (Supplementary Fig. 14b) becomes more symmetric. Moreover, we extracted CL spectra from individual grains located in different regions in the film samples (1 cm² area). Cation homogenization by nano-GT shallowing becomes even more evident over the long-range grain microstructure (Supplementary Figs. 15 and 16). We additionally performed hyperspectral PL imaging, showing consistent findings (Supplementary Fig. 17). We further prepared a perovskite film with an intermediate nano-GT depth (average d_{GT} , 6.8 nm; Supplementary Fig. 18). While δ -CsPbI₃ is not observed in this sample, a modest discrepancy in the CL peak centres for different grains is retained (Fig. 2b,e,h and Supplementary Fig. 19). This observation attests to the d_{GT} dependence of cation homogenization. There is a certain level of CL emission differences in the intensity and peak position between GBs and grain interiors, primarily owing to Cs enrichment near the GBs. This Cs-enriched GB feature is also alleviated with shallower nano-GTs (Supplementary Fig. 20). We also observed that nano-GT shallowing causes

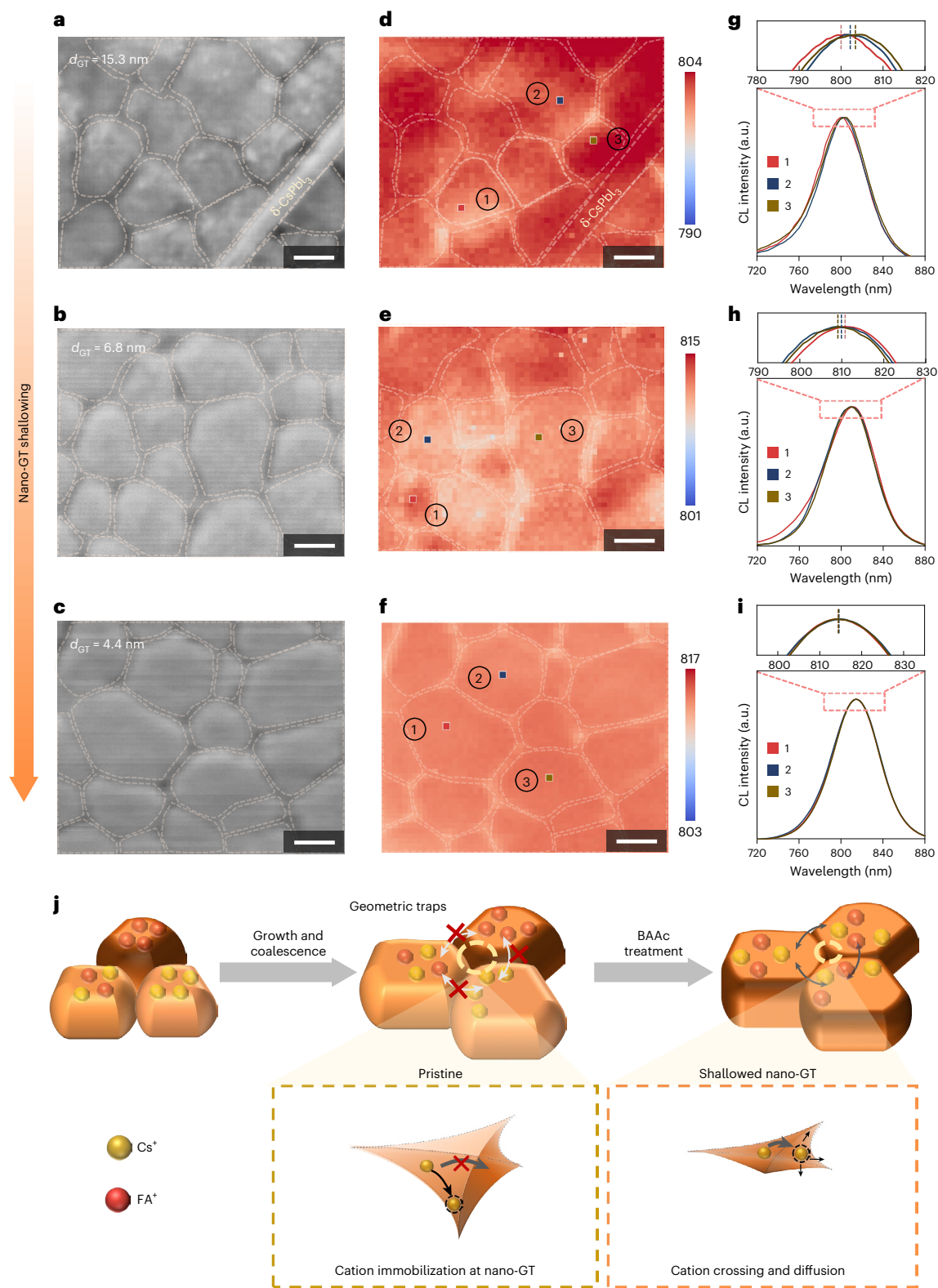


Fig. 2 | Influence of the nano-GT depth on the nanoscale cross-grain cation homogeneity. **a–c**, SEM images of the perovskite films with different average d_{GT} : 15.3 nm (**a**), 6.8 nm (**b**) and 4.4 nm (**c**). **d–f**, CL mapping of the peak emission wavelengths of the perovskite films with different average d_{GT} : 15.3 nm (**d**), 6.8 nm (**e**) and 4.4 nm (**f**). **g–i**, Representative local CL spectra (from spots marked in **d–f**) of the perovskite films with different average d_{GT} : 15.3 nm (**d**), 6.8 nm (**e**) and 4.4 nm (**f**). The rod-shaped grain has been assigned to δ -CsPbI₃. **j**, Schematic illustration showing how the nano-GT depth influences cation

homogeneity in the FA–Cs perovskite thin film and how a shallowed nano-GT improves cation homogeneity. Rapid supersaturation creates perovskite nuclei with cation inhomogeneity, followed by grain growth and meanwhile cation exchange. During the later process, nano-GTs act as geometric traps that can capture migrating cations and cause cation immobilization. Shallowing nano-GTs facilitates cation motion and then attribute more homogeneous cation distribution among adjacent grains.

a small redshift in the CL emission peak, mostly attributed to the BAAC induced compressive strain on the film bottom³⁷.

Figure 2j illustrates the mechanism for shallowed nano-GTs induced cross-grain homogenization. During the crystallization of FA–Cs perovskite films, the supersaturation leads to nucleation and growth of perovskite grains with different FA–Cs cation compositions corresponding to the local environment. The grains will then coalesce, during which cationic interdiffusion will occur to induce composition homogenization (Supplementary Fig. 21). Perovskite in PSCs exhibits surface-dominated microstructure, and thus, the solid-state cationic interdiffusion is expected to be mainly cross multi-grain surfaces³⁸. Nano-GTs, acting like geometrical traps on the flat film landscape, hinder the thermodynamic motions of cations. In Supplementary Fig. 22, we consider various atomistic scenarios of FA/Cs cation movement across various grooving conditions, illustrating the role of nano-GT geometry. Interestingly, the nanoscopic cation homogenization on the film bottom surface also causes a beneficial ‘chain’ effect on the cation distribution on the top surface and across the film thickness (Supplementary Figs. 23–27 and Supplementary Note 3).

Enhanced film properties with shallowed nano-groove traps

In terms of optoelectronic properties, as revealed by DFT calculations (Supplementary Fig. 28), in the pristine film, the grain-to-grain cationic composition difference creates in-plane fluctuations in the energy level of conduction band minimum (CBM), while the valence band maximum (VBM) profile across different grains can remain flat³⁹. This causes an unfavourable injection of photocarriers from Cs-rich to FA-rich grains and thus resulting in thermal loss (Fig. 3a,b). By contrast, the shallowed nano-GT film shows an in-plane homogeneous surface electronic structure. Experimentally, we used Kelvin-probe force microscopy (KPFM) to probe the nanoscale surface potential, indicating that the nanoscopic grain-to-grain inhomogeneity in surface potential is alleviated after nano-GT shallowing (Supplementary Fig. 29). Then we performed transient absorption (TA) spectroscopy on both films. Both bandgap renormalization and hot carrier cooling occur within 2 ps for all samples after photoexcitation⁴⁰. The full-width at half-maximum (FWHM) at the maximum absorption scenarios (1.56 ps) are 29.5 nm and 26.7 nm for the pristine and shallowed nano-GT films, respectively. This indicates that the shallowed nano-GT film tends to exhibit less CBM energy level spreading than the pristine film. Meanwhile, more rapid decay kinetics is observed in the shallowed nano-GT film, indicative of accelerated carrier injection across the perovskite–ETL heterointerface (Fig. 3c and Supplementary Fig. 30). We then performed time-resolved photoluminescence (TRPL) measurements from the bottom surfaces of the pristine and shallowed nano-GT films on quartzes. As shown in Fig. 3e, the average PL lifetime (τ) for the shallowed nano-GT film is 202 ns, larger than that (130 ns) for the pristine film, indicating the suppression of non-radiative recombination⁴¹. We then performed TRPL experiments on perovskite films on ETLs. The PL decay becomes faster ($\tau = 30$ ns) in the shallowed nano-GT film (Fig. 3f), compared with the pristine case ($\tau = 117$ ns). Consistent results from steady-state PL experiments are in Supplementary Fig. 31. Cross-grain cation homogenization also leads to a high spatial uniformity of PL emission intensity in the shallowed nano-GT film (Supplementary Fig. 32).

We examined the effect of cross-grain cation homogenization on the perovskite film stability and reliability. We compared the ultraviolet-visible spectra evolutions under 22 °C and 40% relative humidity (RH) conditions. The shallowed nano-GT film shows superior stability with the absorbance nearly unchanged after 300 h storage, while the absorption characteristics of the pristine film decay fast (Fig. 3g). The X-ray diffraction (XRD) results in Fig. 3h and Supplementary Fig. 33 attest to the phase stability enhancement by showing the suppression of δ -phase formation in the shallowed nano-GT film. We further observed that the degradation is linked to the propagation

of phase transformation from perovskite to δ -phase across the film microstructure (Supplementary Fig. 34). The pinholes formed during the early stage of the degradation are mainly located around the nano-GTs. We consider two factors enhancing stability in the shallowed nano-GT film. First, the grain-to-grain cation homogeneity in the shallowed nano-GT film ensures an ideal Goldschmidt tolerance factor for all perovskite grains, while in the pristine case, many grains deviate from the ideality in terms of the FA–Cs composition and thus tolerance factor. Second, the altered nano-GT geometry may contribute to reduced moisture or solvent trapping in the nanovoids²¹. We then evaluated the perovskite film adhesion under the ASTM D3359 standard⁴² (Fig. 3i). The typical adhesion grade of the shallowed nano-GT film is classified as 4B, higher than the pristine case (2B). This improvement is statistically confirmed in Supplementary Fig. 35. In addition, we have frequently observed the presence of SnO₂ nanoparticles on the surface of the delaminated, shallowed nano-GT film (Supplementary Fig. 36). We attributed this to the formation of a mechanically robust perovskite–ETL heterointerface that allows the pull-out of SnO₂ nanoparticles during interlayer delamination.

Improved solar cell performance by nano-groove trap engineering

We then assessed the impact of nano-GT shallowing induced cross-grain cation homogenization on the device PCE and stabilities. The adopted device structure of PSCs is shown in Supplementary Fig. 37. According to the typical current density–voltage (J – V) curves (Fig. 4a), the shallowed nano-GT PSC shows a higher PCE of 25.62%, with a short-circuit current density (J_{sc}) of 25.7 mA cm^{−2}, an open-circuit voltage (V_{oc}) of 1.22 V, and a fill factor (FF) of 0.817. We sent one shallowed nano-GT PSC to an independent CNAS-accredited test centre, which delivers a PCE of 25.24% (Supplementary Fig. 38). For comparison, the pristine PSC only shows a PCE of 23.54% with a V_{oc} of 1.19 V, aJ_{sc} of 25.3 mA cm^{−2} and an FF of 0.782. One important contributing factor to the PCE improvement is the V_{oc} increase. Most of the shallowed nano-GT PSCs (39 out of 40) demonstrate high V_{oc} values larger than 1.20 V (Fig. 4b), which can be linked to the ideally flat CBM of the film surface (Fig. 3b), resulting from cross-grain cation homogenization. We also monitored the stabilizing photocurrent outputs at the maximum power points (MPPs), which showed the higher PCE and faster photoelectric response in the shallowed nano-GT PSC (Fig. 4c). We further statistically confirmed the excellent reproducibility of the observed J – V performance improvement (Supplementary Fig. 39).

The shallowed nano-GT PSC demonstrates higher stabilities in a range of standardized ageing conditions. This is because Spiro-OMeTAD hole-transporting layer (HTL) was based on molecular packing that can be fatigued during long-time working conditions, which may also affect the device durability results^{43,44}. Therefore, we additionally fabricated PSC devices with poly(bis(4-phenyl)(2,4,6-trimethylphenyl)amine) (PTAA) for stability tests. We first conducted the MPP tracking tests with the ISOS-L-1I protocol. As shown in Fig. 4d, the shallowed nano-GT PSC demonstrates an impressive 90.2% PCE retention upon 1,000 h continuous one-sun-intensity light illumination, surpassing the pristine case (80.0% retention upon 170 h). Herein the typical initial PCEs of pristine and shallowed nano-GT PSCs were 20.14% and 21.20%, respectively. In the absence of an MPP biasing, an even higher PCE retention (92.0%) can be achieved in the shallowed nano-GT case (Supplementary Fig. 40). Then, we performed the thermal cycling (MQT11) and damp heat (MQT13) tests under the IEC 61215:2021 standard, the results from which can be mainly linked to the thermo-mechanical durability and phase stability of the perovskite layer, respectively⁴⁵. For the thermal cycling test, the shallowed nano-GT PSC retains 95.7% of the initial PCE upon 300 thermal cycles, higher than that (74.8%) for the pristine PSC (Fig. 4e). Herein the typical initial PCEs of pristine and shallowed nano-GT PSCs were 20.16% and 21.00%, respectively. For the damp heat tests, after 1,000 h, a 90.5% PCE retention is achieved

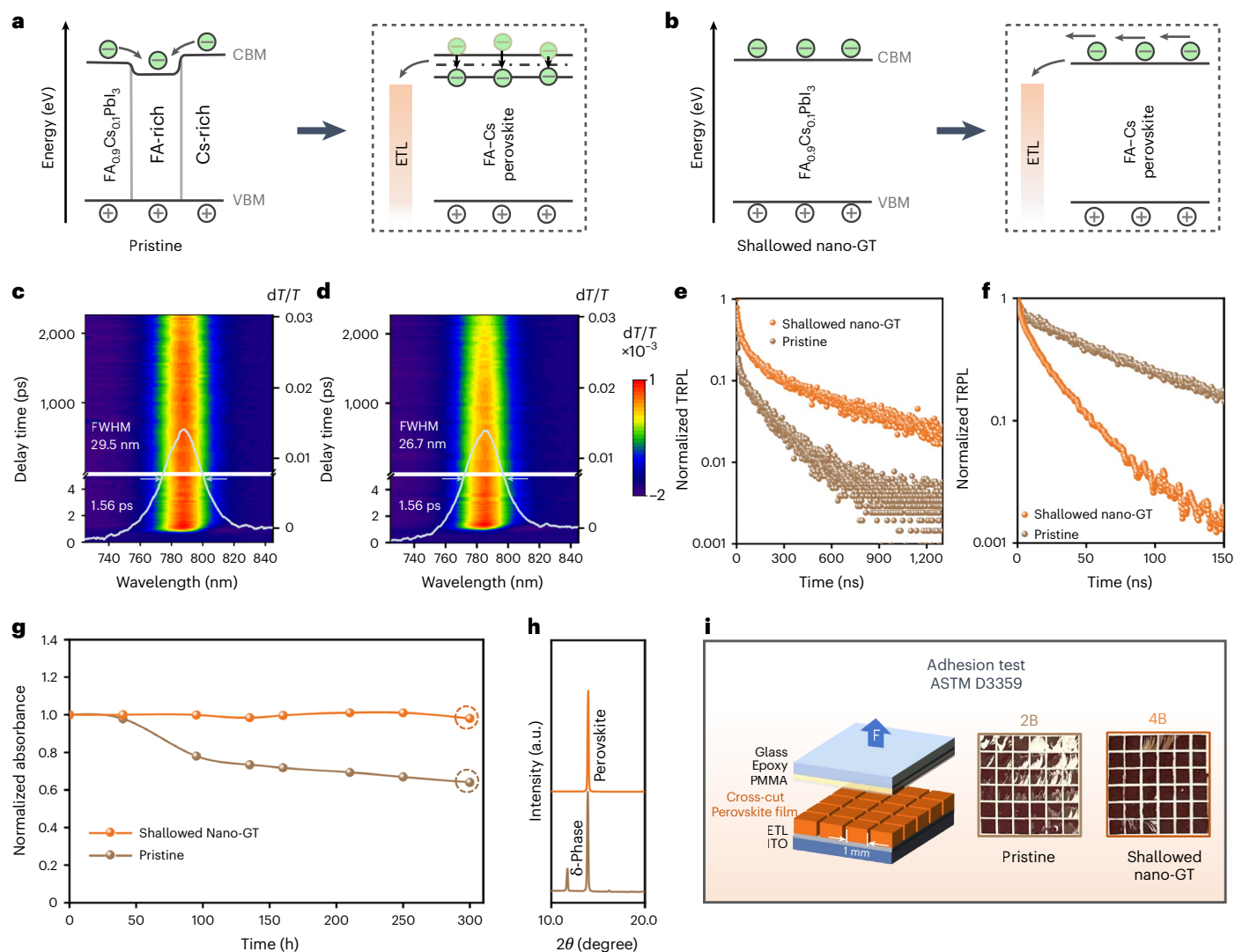


Fig. 3 | Influence of cross-grain cation homogeneity on film properties. **a, b**, Lateral energy level alignment of the surface featuring grains with different FA–Cs compositions, for example, $\text{FA}_{0.9}\text{Cs}_{0.1}\text{PbI}_3$, FA-rich ($\text{FA}_{1-x}\text{Cs}_x\text{PbI}_3$ ($x < 0.1$)), Cs-rich ($\text{FA}_{1-x}\text{Cs}_x\text{PbI}_3$ ($x > 0.1$)), for the pristine (**a**) and shallowed nano-GT (**b**) films, and the resultant energetics across the perovskite–ETL interface: the pristine sample exhibits localized variation in the band edges owing to inhomogeneous cation distribution, whereas the target sample demonstrates uniform band edge. **c, d**, TA maps of pristine (**c**) and shallowed nano-GT (**d**) perovskite films (with ETL), respectively. The overlapped curves are the TA spectra at a delay time of 1.56 ps. **e, f**, TRPL spectra of the pristine (**e**) and shallowed nano-GT (**f**) perovskite films without and with ETL, respectively. The fitting parameters in Supplementary Tables 1 and 2. **g**, Absorbance (at 760 nm)

variations of the pristine and shallowed nano-GT perovskite films as a function of storage duration in the laboratory ambient environment. **h**, XRD patterns of the pristine and shallowed nano-GT perovskite films after 300 h (corresponding to the samples with dash circles in **g**, respectively) storage in the ambient conditions. **i**, Schematic illustrations of the film adhesion test (ASTM D3359) for qualitatively evaluating the mechanical reliability of the buried perovskite–ETL heterointerface. Inset: optical photos (each cross-cut square is 1 mm²) of residual perovskite films with the degree of perovskite layer removal. The degree of perovskite layer removal is graded using a standardized scale, ranging from 0B to 5B, with 5B indicating no adhesion loss and 0B indicating complete removal. FWHM, full-width at half-maximum.

for the shallowed nano-GT PSC, compared with 42.8% for the pristine case (Fig. 4g). Herein the typical initial PCEs of pristine and shallowed nano-GT PSCs were 20.19% and 21.90%, respectively. Overall, the enhancement of device durability parameters in different standardized tests is irrespective of the HTL type used, attesting to the core role of the perovskite layer. For example, when using the Spiro-OMeTAD HTL, nano-GT shallowing increases the PCE retention of PSCs upon 195 thermal cycles from 23.7% to 82.3% (Supplementary Fig. 41). In Fig. 4f, h, we illustrated the materials mechanisms linked to observed device stability enhancements. In thermal cycling tests, the mechanical integrity of the perovskite–ETL heterointerface is essential. In the pristine PSCs, these regular nano-GTs and buried δ -CsPbI₃ rod-like crystals may serve as inherent, buried crack nuclei that will trigger

interfacial delamination and its propagation during the test. Once the nano-GT depths are reduced, these crack nuclei are eliminated, reducing the occurrence of detrimental film delamination. In damp heat tests, cross-grain cation homogenization, induced by nano-GT shallowing, creates individual perovskite grains nearly all in the ideal composition and Goldschmidt's tolerance factors, unlocking higher phase stability under light and device-operation conditions.

Conclusion and outlook

We have revealed the non-trivial grain-to-grain cation inhomogeneity in the pristine FA–Cs mixed-cation perovskite film. We then established an intriguing relation between such grain-to-grain variations and nano-GT geometries. Regardless of their relatively low dimensions,

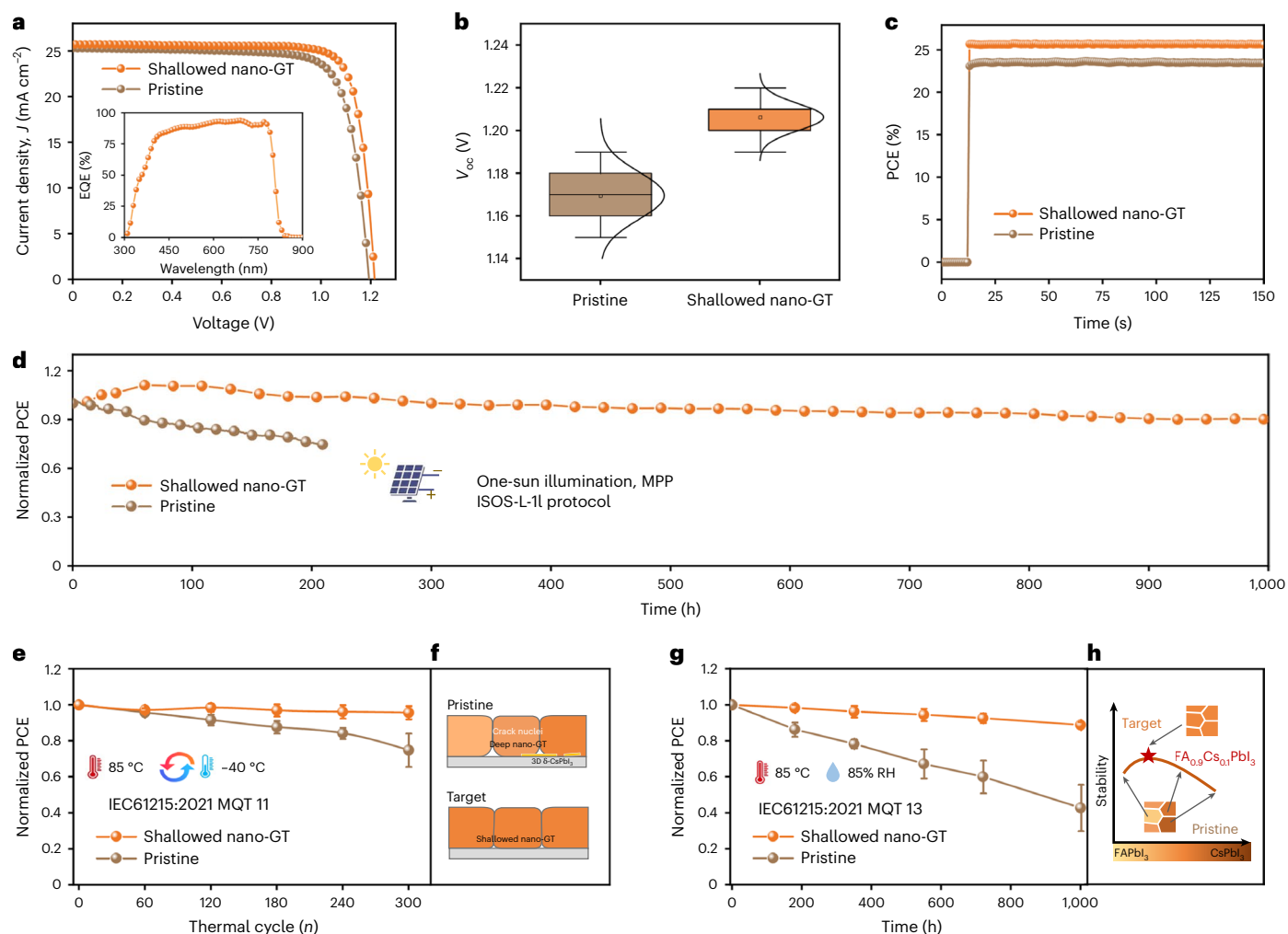


Fig. 4 | PSC performance with cross-grain cation homogenization. **a**, $J-V$ curves scanned from V_{OC} to J_{SC} . Inset of **a** is the corresponding EQE spectrum of PSCs with the shallowed nano-GT film. **b**, Statistics of extracted V_{OC} from $J-V$ scans for both PSCs (sample size $n = 40$). **c**, Corresponding current outputs at the MPPs for both PSCs. The box plot in **b** shows the mean, median line, 25–75% box limits with $1.5\times$ interquartile range whiskers. **d**, MPP tracking stability test of PSCs with the pristine and shallowed nano-GT perovskite films based on the ISOS-L-1I protocol. **e**, Thermal cycling stability test result of PSCs with the pristine and shallowed nano-GT perovskite films based on the IEC 61215:2021 MQT11 protocol (sample size $n = 5$ for both cases). **f**, The pristine PSC contains

two types of heterointerfacial microstructures, regular (deep) nano-GTs and 3D δ -CsPbI₃, which can serve as crack nuclei. This leads to weak adhesion and reduced durability upon thermal cycling, while the PSC with shallowed nano-GTs exhibits much elevated integrity at the perovskite heterointerface. **g**, Damp-heat test under 85% RH and 85 °C heating based on the IEC 61215:2021 MQT13 protocol (sample size $n = 5$ for both cases). **h**, The PSC with shallowed nano-GTs contains numerous individual perovskite grains with FA–Cs compositions deviating from the ideal prescribed composition, responsible for an overall slower decay of device PCE. Error bars in **e** and **g** represent the standard deviation of the measurements obtained from five samples in both cases.

nano-GTs influence the FA–Cs cation landscape of the entire perovskite film. Nano-GT shallowing facilitates grain-to-grain cation homogenization, leading to improved interfacial properties and, correspondingly, better solar cells. This work highlights the vital importance of nanoscopic grain-resolved microstructures in perovskites, which can be beyond the nano-GT. Fundamental research along this line offers a new perspective for improving PSCs, complementing the existing strategies. Beyond the FA–Cs system, the nanoscopic cross-grain compositional engineering demonstrated in this work may also be extended to improving metal cation and halide anion homogeneities for these mixed Pb–Sn and mixed I–Br perovskites, which are essential for tandem device developments⁴⁶. Beyond photovoltaics, resolving nanoscopic grain-to-grain structural and functional inhomogeneities in perovskites can impact the development of highly sensitive photodetectors and high-colour-purity light-emitting devices. In the broad context of nano research, this work represents a prototype for resolving surface nano-morphologies for homogeneous properties among the vast family of multi-elemental nanomaterials.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41565-025-01854-y>.

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Methods

Raw chemicals

PbBr₂ (≥99%), 4-*tert*-butylpyridine (TBP, 96%), bis(trifluoromethane)sulfonimide lithium salt (99.95%), poly(methyl methacrylate) (PMMA), formamidine acetate (FAAc, 99%), dimethylformamide (DMF, 99.8%), dimethyl sulfoxide (DMSO, 99.9%), chlorobenzene (CB, 99.8%), toluene (TB, 99.8%), diethyl ether (DE, 99%), acetone (99.5%) and isopropanol (IPA, ≥99.8%) were acquired from Merck. PbI₂ (>99%) was acquired from TCI. NH₂CH = NH₂Br (FABr, >99.99%), NH₂CH = NH₂I (FAI, >99.99%), CH₃NH₃Cl (MACl, >99.99%), PbBr₂ (≥99%), *n*-butylammonium acetate (BAAc, >99.99%), *n*-butylammonium iodide (BAI, >99.99%), CH₃NH₃I (MAI, >99.99%), CH₃NH₃Br (MABr, >99.99%) and methoxy-phenethylammonium iodide (MeO-PEAI, >99.99%) were purchased from Greatcell Solar. Tin(IV) oxide (SnO₂, 15 wt% in H₂O colloidal dispersion), CsI (99.998%), PTAA (average *M_n* 15,000–20,000) and tetrakis(pentafluorophenyl)borate (TPFB, 99%) were acquired from Alfa Aesar. Spiro-OMeTAD (99.8%) was purchased from Borun Chemical. All raw chemicals were used as received without further purification.

Preparation of solutions and thin films synthesis

For preparing the ETL solution, the acquired SnO₂ nanoparticles were diluted using distilled water five times. For preparing the Spiro-OMeTAD HTL solution, 72.3 mg Spiro-OMeTAD, 28.8 μl TBP and 17.5 μl bis(trifluoromethane)sulfonimide lithium salt acetonitrile solution (520 mg ml⁻¹) were dissolved into 1 ml CB. Then, 50 μl of the solution was spin-coated onto the perovskite layer at 3,000 rpm for 30 s. For preparing the PTAA HTL solution, 30 mg PTAA and 3 mg TPFB were dissolved into 1 ml CB. Then, 50 μl of the solution was spin-coated onto the perovskite layer at 3,000 rpm for 30 s. For preparing the BAAc solution, the BAAc was dissolved in IPA solution. Afterwards, the solution directly was spin-coated onto the SnO₂ surface, followed by annealing at 75 °C for 5 min.

FA_{0.9}Cs_{0.1}PbI₃ perovskite films are used for fundamental investigation of the microstructure of nano-GT. For preparing these perovskite thin films, typically, a perovskite precursor solution with 1 M concentration was prepared by co-dissolving 154.8 mg FAI, 26.0 mg CsI and 461.1 mg PbI₂ in 1 ml mixed solvent of DMF/DMSO (v/v, 7:3). Then, 50 μl of this precursor was spread on the SnO₂ ETL-coated ITO substrates, followed by a 3-stage spin-coating process (500 rpm for 5 s, 1,000 rpm for 10 s and 5,000 rpm for 30 s). During the third stage, 400 μl of TB was dripped at the centre of the spinning substrate. Then, the films were annealed at 150 °C for 10 min to form the final perovskite films. For preparing the film sample with pristine nano-GT depth, no BAAc was used in the ETL fabrication; for preparing the film sample with an intermediate nano-GT depth and shallowest nano-GT depth, the used BAAc concentration was 10 mg ml⁻¹ and 30 mg ml⁻¹ in IPA, respectively.

(FA_{0.95}Cs_{0.05})PbI₃)_{0.975}(MAPbBr₃)_{0.025} perovskite films are used for the device demonstration. For preparing these perovskite thin films, the precursor was first prepared by co-dissolving 705.3 mg of PbI₂, 228.8 mg of FAI, 18.2 mg of CsI, 33.7 mg of MACl and 18.2 mg of MAPbBr₃ single crystal in mixed solvent of 890 μl DMF and 110 μl DMSO. The MAPbBr₃ single crystal was synthesized by mixing MABr and PbBr₂ in a 1:1 molar ratio with a concentration of 0.2 M in DMF solvent and capturing the single crystal with dichloromethane anti-solvent. Then, 50 μl of precursor was spin-coated at 1,000 rpm for 10 s and then at 4,000 rpm for 30 s. In the last 20 s of the second step, the anti-solvent (1 ml of DE) was dropped in the centre of the substrate within 1 s. After the spin-coating was completed, the films were annealed at 100 °C for 40 min. For surface passivation, a solution of 1 mg MeO-PEAI dissolved in 1 ml IPA was spin-coated onto the perovskite films at 4,000 rpm for 30 s.

Materials characterization and property measurement

The surface geometry of nano-GTs was measured using an AFM platform (Multi-mode 8, Bruker) using a 0.01–0.025 Ω cm antimony (*n*-)doped Si

tip (RTESPA-300) in a non-contact tapping mode. The KPFM was performed with a 0.01–0.025 Ω cm antimony (*n*-)doped Si tip (SCM-PIT-V2) in a PeakForce-KPFM mode using an AFM platform (Dimension ICON, Bruker). Time-of-flight secondary ion mass spectrometry depth-profile analysis measurement was performed using an IONTOF M6 instrument (IONTOF GmbH). The SEM images of the thin films were observed using a scanning electron microscope (Gemini 1530, LEO, Zeiss). XRD was performed using a high-resolution diffractometer (D8 Advance, Bruker) with Cu Kα radiation ($\lambda = 1.5406 \text{ \AA}$) in the θ – 2θ scan mode. X-ray photoelectron spectroscopy measurements were performed using a spectroscopic system (Thermo Scientific K-Alpha, Thermo Fisher Scientific). Optical absorption spectra were acquired using a spectrophotometer (Cary 300, Agilent). For the steady-state PL and TRPL measurements, the sample was excited with a 375 nm picosecond laser (LDH-D-C-375, PicoQuant). TRPL spectra were fitted with double exponential decay function: $f(t) = A_1 \exp(-t/\tau_1) + A_2 \exp(-t/\tau_2)$, where A_1 and A_2 are the decay amplitudes, and τ_1 and τ_2 are the decay time constants. The PL mapping images were collected using a laser scanning fluorescence confocal microscope (TCSSP8, Leica) with excitation at 488 nm. The excitation laser was focused onto the samples using a lens. The steady-state PL signal was channelled into a spectrograph (Ando Kymera 328i, Ando Electric) and subsequently captured by an electron-multiplying charge-coupled device (Andor iXon Life 888, Oxford Instruments). For TA measurements, infrared laser pulses were generated using an amplifier (Pharos, Light Conversion) with a pulse duration of 170 fs and a wavelength of 1,030 nm, operating at a repetition rate of 1 kHz. The pump pulse was directed into an optical parametric amplifier (Orpheus-F, Light Conversion) and a pulse laser with a wavelength of 640 nm was generated. The pump power was subsequently attenuated using a neutral density filter wheel and a neutral density filter, resulting in a sample fluence of 2.31 μJ cm⁻². A probe laser carrying 0.1 mJ of pulse energy was attenuated and focused onto an yttrium aluminium garnet (YAG) crystal to generate the supercontinuum probe spectrum. The probe laser is focused onto the sample, overlapping with the pump laser spot. The temporal delay was controlled using a translation stage and the collected signal was analysed using a USB spectrometer. The large-scale PL mapping was performed with MicroRaman/Photoluminescence System (InVia, Renishaw) with excitation at 532 nm.

Methods for film delamination and nano-GT geometric parameter determination

To facilitate the delamination of the perovskite film, sample specimens featuring a multi-layer structure of glass/ITO/ETL/perovskite/PMMA/epoxy/ITO/glass were utilized. Initially, SnO₂ ETL was deposited onto ITO-coated glasses to prepare these specimens. Subsequently, the perovskite layer was applied, followed by spin-coating a PMMA layer using a 10 wt% CB solution at 6,000 rpm for 60 s. The purpose of the PMMA coating was to shield the perovskite from chemical degradation. Then, a thin layer of epoxy (2 μm) was applied to adhere a cover glass onto the film surface. The epoxy underwent curing for 24 h under dry air conditions (with less than 20% RH). Finally, an upward force was applied to the cover glass, causing the perovskite thin film to delaminate from the ETL, allowing for subsequent measurements.

The determination of nano-GT geometric characterization relies on the analysis of AFM images and surface profiles, facilitated by the NanoScope Analysis (v1.8) software. For the measurement of nano-GT depth, the lowest point is determined by analysing the profile passing through the lowest point at the nano-GT, while the highest point is obtained by averaging the highest points of the three grains near the nano-GT. The depth is then calculated by subtracting the lowest value from the highest value. Measurements are taken once on each side, and the average is calculated. Here we only considered the dominating nano-GT features formed between different grains with sizes larger than 80 nm, to exclude the influence of ultrasmall grains occasionally observed in the AFM images and formed by unknown reasons.

Calculation of nano-GT geometric parameters

The relationship with surface energy (γ_s) and groove root line tension energy (γ_{ls}) can be determined⁴⁷:

$$\gamma_{ls} = 2\gamma_s \left(\cos \frac{\theta}{2} - \cos \frac{\xi}{2} \right) \cdot R \quad (2)$$

where R is the radius of curvature at the given point of the groove root and θ is the side angle of GB groove as elaborated elsewhere²⁰. Therefore, at a nano-GT, γ_{ls} and γ_s are positively correlated.

For the nano-GT at triple junctions, the total depth of a nano-GT (d_{GT}) is the sum of the grain-boundary groove depth (d_G) and the groove root depth (d_{GR}), as illustrated in Fig. 1a:

$$d_G = \frac{l_G \cdot \cos \left(\frac{\pi}{2} - \theta \right)}{1 + \sin \left(\frac{\pi}{2} - \theta \right)} \quad (3)$$

$$d_{GR} = \frac{l_{GR} \cdot \cos \left(\frac{\pi}{2} - \zeta \right)}{1 + \sin \left(\frac{\pi}{2} - \zeta \right)} \quad (4)$$

where l_G is the length of the projected length of the groove in the vertical direction and l_{GR} is the length of the projected length of the groove in the vertical direction. Supplementary Fig. 43. comprehensively illustrates d_G and d_{GR} relation with θ and ζ .

Therefore,

$$d_{GT} = d_G + d_{GR} = \frac{l_G \cdot \cos \left(\frac{\pi}{2} - \theta \right)}{1 + \sin \left(\frac{\pi}{2} - \theta \right)} + \frac{l_{GR} \cdot \cos \left(\frac{\pi}{2} - \zeta \right)}{1 + \sin \left(\frac{\pi}{2} - \zeta \right)} \quad (5)$$

Hyperspectral CL microscopy

The CL measurements were conducted using an FEI Quattro environmental SEM, equipped with a Delmic Sparc CL collection module. This module incorporates a parabolic mirror to efficiently gather CL signals emitted from the film upon electron-beam excitation. The sample was excited using an electron beam with an acceleration voltage of 5 kV, accompanied by a beam current of 14 pA, directed through a perforation in the parabolic mirror. CL signal collection was conducted with an acquisition time of 200 ms per spectrum. The pixel size was set to 30 nm. All measurements were carried out in a low-vacuum environment with 50 Pa H₂O vapour, effectively mitigating sample charging. Notably, it was observed that these measurement conditions did not result in any discernible sample degradation throughout the measurement process.

Spatially resolved CL emission peak position maps of the perovskite films were achieved through automated emission fitting with the GaussFit function implemented in Python. The formula for the entire model would be the sum of the Gaussian functions along with the constant:

$$f(x) = c + \sum_{i=1}^5 A_i \cdot \exp \left(-\frac{(x - \mu_i)^2}{2\sigma_i^2} \right) \quad (6)$$

where c is the constant background level, A_i , μ_i , σ_i and the amplitude, centre and standard deviation of the i th Gaussian peak, respectively. This composite function represents the Gaussian mixture model with five Gaussian peaks fitted to the input data.

Adhesion test

The adhesion/delamination classification is determined with ASTM D3359. The film sample specimens are made with a multi-layer structure of glass/ITO/SnO₂(ETL)/perovskite/PMMA/epoxy/glass. A cross-hatch pattern with 1 mm² individual partition was first made on the perovskite film surface. The cuts penetrated through the perovskite layer, reaching

the underlying ETL layers. Then we applied adhesive epoxy firmly onto the cross-cut area, exerting pressure to promote intimate contact between the epoxy and the multi-layered film. By applying a constant force (F) from the film, the perovskite film is gradually delaminated from the ETL-coated ITO glass. Via observing the retained parts of the cross-cut area, the adhesion of the film can be classified from 0B to 5B, with 5B indicating no adhesion loss and 0B indicating complete failure.

DFT calculation

Relative energy of the perovskite–SnO₂ interface. The first-principles computations are performed using Vienna Ab initio Simulation Package (VASP 5.4). An energy cut-off of 520 eV is used. Structural geometry optimization was performed using Perdew–Burke–Ernzerhof exchange–correlation functional and with energy convergence and force convergence of 10^{−6} eV and 0.02 eV Å^{−1}, respectively. Grimme's DFT-D3 correction is adopted to describe the long-range van der Waals interaction. A Γ -centred ($2 \times 2 \times 1$) k -point mesh was used to sample the Brillouin zone. The ion cores are described by using the projector augmented wave method. We used a $2\sqrt{2} \times 2\sqrt{2}$ supercell of (001)-plane SnO₂ surfaces and a 2×2 supercell of (001)-plane FAPbI₃. The slab consists of five symmetric layers of SnO₂ and four of FAPbI₃ with BA and Ac ions at the interface, with a vacuum size of about 20 Å.

Bandgaps of FA_{1-x}Cs_xPbI₃ ($x = 0.08, 0.17, 0.25$). Projector augmented wave pseudopotentials as implemented in the VASP code were used to carry out the DFT calculations. The generalized gradient approximation in Perdew–Burke–Ernzerhof parameterization was used to describe the electron-ion and exchange correlation potentials. The energy cut-off was set to be 450 eV, and the total energies and forces were calculated with a precision of 10^{−3} eV and −0.02 eV Å^{−1} for electronic and force convergence, respectively. The Γ -centred k -meshes with the k -spacing of 0.2 Å^{−1} was used to sample the Brillouin zone for the calculation. To model the FA_{1-x}Cs_xPbI₃ perovskites, a $2 \times 2 \times 3$ supercell containing 12 formula units of FAPbI₃ was built.

Device fabrication and testing

ITO glass substrates were washed thoroughly, involving sequential ultrasonic cleaning with cleaning fluid, deionized water, acetone and IPA, with each step lasting 20 min. The SnO₂ layer was spin-coated onto the ITO glass substrate at 3,000 rpm for 30 s and annealed at 150 °C for 30 min in ambient air. The ETL-coated substrate was then transferred to a nitrogen-filled glovebox (H₂O < 0.01 ppm, O₂ < 0.01 ppm). The perovskite layer was then deposited using the method mentioned earlier. Following the perovskite deposition, the HTL was applied on top of the perovskite layer through spin-coating at 4,000 rpm for 30 s. Finally, an 80 nm Au electrode was thermally evaporated onto the HTL. The J – V characterization for PSCs was measured by a source meter (2612, Keithley) under AM 1.5 G one-sun illumination (100 mW cm^{−2}) generated by a class AAA solar simulator (Sirius-SS, Zolix) in a nitrogen-filled glovebox. A metal mask made with high precision was used to define the active area for every measurement. The reference cell for calibrating the solar simulator before J – V measurements was an Oriel reference solar cell accredited by NIST to the ISO-17025 standard. External quantum efficiency (EQE) spectra were recorded at a chopping frequency of 165 Hz in AC mode on a solar cell quantum efficiency measurement system (QE-R3011, Enlitech) to ensure the reliability of J – V determined short-circuit densities of our best-performing devices. The scans were performed in the voltage range of −0.02 V to 1.24 V at the reverse direction (from V_{oc} to J_{sc}) with a step size of 0.02 V and a dwell time of 10 ms. Steady-state current and PCE outputs were measured using the 2612 source meter (Keithley) at the voltages determined from the MPPs of the reverse-scan J – V curves.

For the light-soaking stability test, unencapsulated PSCs were placed in a sealed quartz cell holder in an N₂-filled glovebox under a one-sun-intensity illumination generated using white light-emitting

diodes. The operational stability was tested under continuous illumination and MPP tracking. For the thermal cycling stability test, the devices were placed at -40°C to 85°C environments with a ramp rate of $-80^{\circ}\text{C h}^{-1}$ cycled environment control by a programmable temperature chamber. For damp heat stability test, the devices were placed in the climate-controlled chamber with 85°C and 85% RH.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data that support the main findings are available in the main text and the Supplementary Information. Source data are provided with this paper.

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47. Zhao, B., Ziemons, A., Shvindlerman, L. S. & Gottstein, G. Surface topography and energy of grain boundary triple junctions in copper tricrystals. *Acta Mater.* **60**, 811–818 (2012).

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Author contributions

Y.Z. directed the project and conceived the idea. Y.Z. and M.H. co-designed the experiments. Y.Z. and M.A. supervised the project. M.H. performed the materials synthesis, device fabrication and materials/device characterizations. J.Y. and B.J.L. assisted with the CL testing and data analyses. W.Y. assisted in the solar cell fabrication and testing. Pengfei Guo (HKUST) assisted with the device characterization. T.X. assisted with the device stability tests. T.D. performed the XRD measurement and assisted with the mechanical adhesion test. X.Z. performed the theoretical calculations. L.C. and Peijun Guo (Yale) performed the photoluminescence experiments. Y.X. assisted in illustrating the mechanism of cation diffusion between adjacent grains. Y.Z. and M.H. co-drafted the paper, and all co-authors reviewed the paper.

Competing interests

The authors declare no competing interests.

Additional information

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► Experimental design

Please check the following details are reported in the manuscript, and provide a brief description or explanation where applicable.

1. Dimensions

Area of the tested solar cells

☒ Yes
☐ No

We provided the information in the Section of device fabrication and testing. The active area was 0.05 cm².

Explain why this information is not reported/not relevant.

Method used to determine the device area

☒ Yes
☐ No

We provided the information in the Section of device fabrication and testing. The active area was defined using a mask.

Explain why this information is not reported/not relevant.

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction

☐ Yes
☒ No

We experimentally measured the devices in both directions, while herein we discussed only the reverse-scan plots because device hysteresis is typically not notable and it is not a topic of focus.

Voltage scan conditions

☒ Yes
☐ No

We provided the information in the Section of device fabrication and testing. The voltage scan is between 1.24 V to -0.02 V with a step size of 0.02 V and a delay time of 10 ms.

Explain why this information is not reported/not relevant.

Test environment

☒ Yes
☐ No

We provided the information in the Section of device fabrication and testing. All devices were tested in a nitrogen-filled glovebox at room temperature.

Explain why this information is not reported/not relevant.

Protocol for preconditioning of the device before its characterization

☐ Yes
☒ No

Provide a description of the protocol.

We did not apply any specific preconditioning protocol before the device characterization.

Stability of the J-V characteristic

☒ Yes
☐ No

We provided the information in Figure 4c-d. We verified the stability of the J-V characteristics by monitoring the current output of the device at the maximum-power-point voltage.

Explain why this information is not reported/not relevant.

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization

☐ Yes
☒ No

Provide a description of hysteresis or any other unusual behaviour observed during the characterization.

We found device hysteresis is typically not notable and it hysteresis is not a topic of focus.

Related experimental data

☒ Yes
☐ No

We could provide the typical hysteresis plot upon requests.

Explain why this information is not reported/not relevant.

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)

☒ Yes
☐ No

We provided the information in Figure 4a and the measurement method in the Section of device fabrication and testing. External quantum efficiency spectra were recorded at a chopping frequency of 165 Hz in AC mode on a solar cell quantum efficiency measurement system (QE-R3011, Enlitech, China).

Explain why this information is not reported/not relevant.

A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator	☒ Yes ☐ No	We experimentally acquired the integrated JSC from the EQE spectrum is 24.6 mA/cm ² , which compares favorably with the measured JSC of optimal device (25.7 mA/cm ²). <i>Explain why this information is not reported/not relevant.</i>
For tandem solar cells, the bias illumination and bias voltage used for each subcell	☐ Yes ☒ No	<i>Provide a description of the measurement conditions.</i> No tandem solar cells were reported.

5. Calibration

Light source and reference cell or sensor used for the characterization	☒ Yes ☐ No	We provided the information in the Section of device fabrication and testing. The J-V characterization was measured by a source meter (2612, Keithley, USA) under an AM 1.5G spectrum (one-sun illumination; 100 mW/cm ²) generated by a solar simulator (Sirius-SS, Zolix, China). <i>Explain why this information is not reported/not relevant.</i>
Confirmation that the reference cell was calibrated and certified	☒ Yes ☐ No	We provided the information in the Section of device fabrication and testing. The reference cell was an Oriel® reference solar cell accredited by NIST to the ISO-17025 standard. <i>Explain why this information is not reported/not relevant.</i>
Calculation of spectral mismatch between the reference cell and the devices under test	☐ Yes ☒ No	<i>Provide a value of the spectral mismatch and/or a description of how it has been taken into account in the measurements.</i> We did not calculate the spectral mismatch between the reference cell and the devices under test because our current level of solar cell performance measurements is sufficient for verifying the efficiency improvement based on our new method.

6. Mask/aperture

Size of the mask/aperture used during testing	☒ Yes ☐ No	We provided the information in the Section of device fabrication and testing. For the laboratory tests, the typical size of the mask is 0.05 cm ² . <i>Explain why this information is not reported/not relevant.</i>
Variation of the measured short-circuit current density with the mask/aperture area	☐ Yes ☒ No	<i>Report the difference in the short-circuit current density values measured with the mask and aperture area.</i> We did not study the variation of short-circuit current density as function of the mask area, because this is not relevant to the scope of this study that focuses on the science of grain microstructure.

7. Performance certification

Identity of the independent certification laboratory that confirmed the photovoltaic performance	☒ Yes ☐ No	We sent one of our best-performing devices (with shallowed nano-GT) to the Photovoltaic and Wind Power Systems Quality Test Center, Institute of Electrical Engineering, Chinese Academy of Sciences for independent certification. Per suggestion of the test center, the verified device exhibits an active area of 6.937 mm ² (defined using a metal mask), slightly larger than that for our laboratory tested devices. The area of the used mask is verified by National Institute of Metrology of China. Based on these, we acquired a certified PCE of 25.24%. <i>Explain why this information is not reported/not relevant.</i>
A copy of any certificate(s)	☒ Yes ☐ No	We included the copy of certificates in Supplementary Fig. 37. <i>Explain why this information is not reported/not relevant.</i>

8. Statistics

Number of solar cells tested	☒ Yes ☐ No	We provided the information in Figure 4b and Supplementary Fig. 38. The number of tested devices were 40. <i>Explain why this information is not reported/not relevant.</i>
Statistical analysis of the device performance	☒ Yes ☐ No	We provided the information in Figure 4b and Supplementary Fig. 38. <i>Explain why this information is not reported/not relevant.</i>

9. Long-term stability analysis

Type of analysis, bias conditions and environmental conditions

☒ Yes

☐ No

We provided the information in Figure 4d-f and Supplementary Fig. 39-40. We followed the IEC 61215:2021 MQT11, IEC 61215:2021 MQT13, ISOS-L-2I, and ISOS-L-1I protocols to test the long-term stability.

Explain why this information is not reported/not relevant.